ECO 344 - Applied Econometrics I
Chapter 7

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November 2, 2026



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Introduction

Qualitative Information

- Qualitative information is information that cannot be described naturally in numbers. For example: gender, race, religion etc.
- Have you used qualitative information in any analysis using numbers?
- What does it mean when we say that it cannot be represented by numbers? Number vs Category.
- Similarly, some variables are represented by numbers but are not numerical in nature. Cardinal vs Ordinal.
- However, we need to convert categories and ordinal variables into numbers to include them in our analysis. We can include such information using:
  - Dummy variables - They can take only two values (1 and 0).
  - Categorical variables - They can take multiple values but are neither cardinal nor ordinal in nature.
  - Ordinal variables - They can take multiple values that are ordered but not cardinal.
- Be careful while including such variables in the analysis and while interpreting them.

Dummy Variable as an Independent Variable

Dummy Variable as an Independent Variable

- A dummy variable takes a value of either 1 or 0. For example: owning a laptop is a dummy variable which takes the value 1 if the participant owns a laptop and 0 otherwise.
- Similarly, a categorical variable with only two categories can easily be converted into a dummy variable.
  - For example: assume gender has two categories, male & female. We can convert it into a dummy variable.
  - The variable "male" takes the value 1 if the participant is male and 0 if the participant is female.
  - Similarly, the variable "female" takes the value 1 if the participant is female and 0 if the participant is male.
- We can include the dummy variable as an independent variable in the regression. Let us see what this does to our model.

Dummy Variable as an Independent Variable

- Consider the following model:

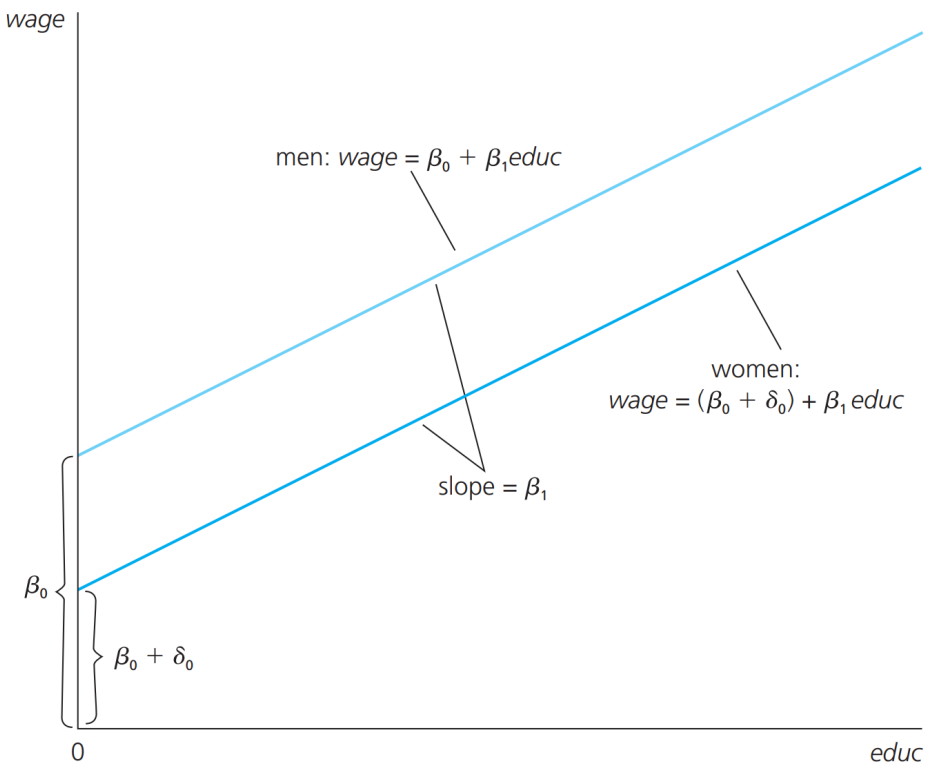
$$wage = \beta_0 + \delta_f female + \beta_1 education + \mu$$

- What would be the interpretation of δ_f in our model?
- To clarify the interpretation of δ_f, let us consider male and female separately.
  - In case of male, $female = 0$. Therefore, $wage = \beta_0 + \beta_1 education + \mu$
  - In case of female, $female = 1$. Therefore, $wage = (\beta_0 + \delta_f) + \beta_1 education + \mu$
  - Holding education fixed, $\delta_f = E[wage|female, education] - E[wage|male, education]$
- So, the coefficient of a dummy variable is the difference in the average outcome between the group coded 1 and the base group (coded 0), keeping the other variables constant.
- In the example above, δ_f is the difference in the average wage of a female and a male with the same education.
- If $\delta_f < 0$, females earn less than comparable males.
- Under what condition can we say that there is no gap between male and female wages? ($\delta_f = 0$)

Dummy Variable: Intercept Change

- Can we call this an intercept change?
- The intercept of male is β_0 while the intercept of female is $\beta_0 + \delta_f$. The slope of education is the same for both groups.

FIGURE 7.1 Graph of $wage = \beta_0 + \delta_0\, female + \beta_1\, educ$ for $\delta_0 < 0$.



Dummy Variable Trap

- Can we add both the male and the female dummy variable in the regression?
- **NO**, because of perfect multicollinearity as $male + female = 1$, which is the constant. This is the Dummy Variable Trap.
- The only way to add both variables is to run the regression without an intercept.
  - The benefit is that the coefficients are directly the intercepts of male and female.
  - However, there is no agreed upon R-Squared calculation for a regression without an intercept, and testing the difference between groups becomes harder.
- Generally, the coefficient of a dummy variable tells us the difference in y as compared to the base group (the group where the dummy variable is zero).
- The choice of base group does not change the fit of the model. It only changes how the coefficients are read.

Is the Gap Discrimination?

- Now, can we say based on the model discussed before that δ_f really measures discrimination against females?
- Think about the ceteris paribus interpretation of δ_f and the model together.
- δ_f is the gap that remains after controlling for the variables in the model. It is consistent with discrimination.
- But it also captures any productivity differences between males and females that we have not controlled for, such as experience, tenure or occupation.
- So the dummy gives us a conditional gap. A causal claim about discrimination needs more than adding a dummy to the regression.

Independent Dummy Variable with Dependent Log Variable

- How do we interpret the coefficient of a dummy variable when the dependent variable is $\log(y)$?

$$\log(wage) = \beta_0 + \delta_f female + \beta_1 education + \mu$$

- The coefficient has an interpretation in terms of percentage.
- The approximate difference is $100 \times \delta_f$ percent. A female earns about $100 \times \delta_f$ percent less than a male (if $\delta_f < 0$) keeping education constant.
- This approximation is good only when δ_f is small. When the coefficient is large, the log approximation is poor.
- The exact percentage difference between the group coded 1 and the base group is $100 \times (e^{\delta_f} - 1)\%$.

Example 1: Hourly Wage Equation (Example 7.1)

- Estimating the model we have been discussing, with experience and tenure added as controls, gets us the following sample regression function.

$$\widehat{wage} = -1.57 - 1.81 female + 0.572 education + 0.025 exper + 0.141 tenure$$

- What is the interpretation of the coefficient of female?
- On average, a female earns 1.81 USD per hour less than a male with the same education, experience and tenure.
- Without any controls the raw gap in the same data is larger. Controlling for the other variables shrinks it, but does not remove it.
- Is this an indication of discrimination? Remember the previous slide.

Example 2: Computer Ownership and College GPA (Example 7.2)

- We are interested in estimating the effect of owning a computer on college GPA. We run the following model:

$$colGPA = \beta_0 + \beta_1 PC + \beta_2 hsGPA + \beta_3 ACT + \mu$$

- We get the following estimation:

$$\widehat{colGPA} = 1.26 + 0.157 PC + 0.447 hsGPA + 0.0087 ACT$$

- What is the interpretation of the coefficient of PC?
- Students who own a computer score 0.157 points higher in college GPA than students who do not own a computer, keeping high school GPA and ACT score constant.
- Is this a causal effect of the computer? What might be missing from the model?

Example 3: Log Wage and Gender (Example 7.4)

- Let us consider the following regression for the wage gap between females and males.

$$\log(wage) = \beta_0 + \delta_1 female + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + \beta_4 tenure + \beta_5 tenure^2 + \mu$$

- The estimated regression function is:

$$\widehat{\log(wage)} = 0.417 - 0.297 female + 0.080 educ + 0.029 exper - 0.00058 exper^2 + 0.032 tenure - 0.00059 tenure^2$$

- What is the interpretation of the coefficient of female?
- Approximately, a female earns 29.7 percent less than a male with the same education, experience and tenure.
- The exact difference is $100(e^{-0.297} - 1) = -25.7$. So a female earns about 25.7 percent less than a comparable male.

Using Multiple Dummy Variables as Independent Variables

Adding Categorical Independent Variable

- We have seen how we can incorporate a dummy variable in the regression. What about a categorical variable with more than two categories?
- Suppose we have the categorical variable education with the categories 1: Less than primary/Illiterate, 2: Primary Education, 3: Matriculation, 4: Finished College, 5: Undergraduate, 6: Master's, 7: Doctorate, 8: Post-Doctorate, in the regression below:

$$wage = \beta_0 + \beta_1 education + \mu$$

- What is the interpretation of β_1 in this case?
- Does the interpretation feel strange? It assumes that every step up the code has the same effect on wage.
- The solution is to create a dummy variable for each category and include them in the regression.
- The model could be:

$$\begin{aligned} wage = \beta_0 &+ \beta_1 illit + \beta_2 prim + \beta_3 matric + \beta_4 college \\ &+ \beta_5 undergrad + \beta_6 master + \beta_7 phd + \beta_8 postdoc + \mu \end{aligned}$$

- What is wrong with the above model?

Adding Categorical Independent Variable

- **Dummy Variable Trap** - the nine terms (intercept plus eight dummies) are perfectly collinear. What is the solution? Drop one of the dummy variables.
- The correct model would be:

$$\begin{aligned} wage = \beta_0 + \beta_1 prim + \beta_2 matric + \beta_3 college + \beta_4 undergrad \\ + \beta_5 master + \beta_6 phd + \beta_7 postdoc + \mu \end{aligned}$$

- The dummy variable we drop acts as the base group or comparison group.
- What is the interpretation of β_5?
- An individual with a master's degree earns β_5 USD per hour more **than an individual with less than primary education**, keeping everything else constant.
- All the discussion regarding a log dependent variable applies here as well.
- To compare two included groups, for example master's and undergraduate, look at $\beta_5 - \beta_4$. A t-test on this difference needs the standard error of the difference, or we can simply change the base group.

Adding Ordinal Independent Variable

- An ordinal variable preserves the order of categories. For example: credit rating of a bond.
- Let us assume the credit rating of a bond CR is coded 4: Excellent, 3: Good, 2: Fair, 1: Poor.
- Consider the following model, which captures the effect of rating on the interest rate.

$$interest = \beta_0 + \beta_1 CR + \mu$$

- What is the interpretation of β_1?
- We can say that β_1 is the change in the interest rate from a 1 unit increase in rating.
- The interpretation looks fine, but it is not right. It forces the move from Poor to Fair to have the same effect as the move from Good to Excellent.
- The solution is to include a dummy variable for each category, leaving one as the base group.

$$interest = \beta_0 + \beta_1 Fair + \beta_2 Good + \beta_3 Excellent + \mu$$

- Here the base group is Poor, and β_3 is the difference in the interest rate between an Excellent and a Poor rating.

Adjusting Categories

- So far, we have seen that categories can be added to the regression through individual dummy variables.
- However, what if we have a large number of categories? For example: the name of the political party.
- We need to add $n-1$ dummy variables to capture the effect of the political party on our outcome. That $n-1$ could be a very large number.
- The solution is to re-categorize into a few dominant parties and put all the rest into "others".
- This reduces the number of dummy variables that need to be included in the regression.
- In this example, we drop the dummy of "others" so that it becomes the base group. The coefficient of each party is then read against all the small parties.

Interaction Terms for Dummy Variables

Dummy Interaction Terms

- We have seen that we can add interaction terms when we think that the effect of one independent variable depends on another.
- We can use the same concept with dummy variables. There are two types of interactions that we can implement in a model involving dummy variables.
  1. Interaction between dummy variables - creates subcategories.
  2. Interaction of a dummy variable and a numerical variable - slope change among different groups.
- Consider the following model:

$$wage = \beta_0 + \beta_1 female + \beta_2 married + \mu$$

- What do β_1 and β_2 represent?
- This model forces the effect of marriage to be the same for males and females. What if we want to see the effect of being a married female?

Interaction Between Dummy Variables

- To capture the effect of being a married female, let us consider the following model:

$$wage = \beta_0 + \beta_1 marriedfemale + \mu$$

  where *marriedfemale* is a dummy variable taking the value 1 when the respondent is female and married.
- What is wrong with the model? Think about the interpretation of β_1.
- The base group now mixes unmarried males, married males and unmarried females. So β_1 compares married females with an average of three different groups.
- The correct model would be:

$$wage = \beta_0 + \beta_1 female + \beta_2 married + \beta_3 female \times married + \mu$$

- Now the base group is clear: unmarried males. Each of the four groups has its own average wage. How? Write the possibilities on paper.

Example: Wage, Gender and Marital Status

- The estimated model of wage on female, married and their interaction is:

$$\widehat{wage} = 5.16 - 0.56\, female + 2.82\, married - 2.86\, female \times married$$

- When female $= 0$ & married $= 0$
  - $\widehat{wage} = 5.16$
  - So β_0 is the average wage of unmarried males.
- When female $= 1$ & married $= 0$
  - $\widehat{wage} = 5.16 - 0.56 = 4.60$
  - So β_1 is the extra/lower wage of an unmarried female relative to an unmarried male.
- When female $= 0$ & married $= 1$
  - $\widehat{wage} = 5.16 + 2.82 = 7.98$
  - So β_2 is the extra/lower wage of a married male relative to an unmarried male.
- When female $= 1$ & married $= 1$
  - $\widehat{wage} = 5.16 - 0.56 + 2.82 - 2.86 = 4.56$
  - So β_3 is the extra/lower wage of a married female beyond the two individual effects.

Allowing for Different Slope

- So far, we have seen how belonging to one category can have an extra/lower effect on the outcome variable.
- What if we want to check whether belonging to a category changes the effect of an independent variable on the outcome?
- For example: does an extra year of schooling have a different effect on the wages of females as compared to males?
- As our question suggests that the effect of education depends on gender, we can include the interaction term of gender and education.
- This allows a different slope between education and wage for males and females.
- It is not necessary that the slope is different for males and females. How would that show up in our estimation?

Allowing for Different Slope

- Consider the following model:

$$wage = \beta_0 + \delta_0 female + \beta_1 education + \delta_1 female \times education + \mu$$

- When female $= 0$

$$wage = \beta_0 + \beta_1 education + \mu$$

  - Therefore, β_1 is the effect of education on the wages of males.
  - β_0 is the average wage of a male with zero education.

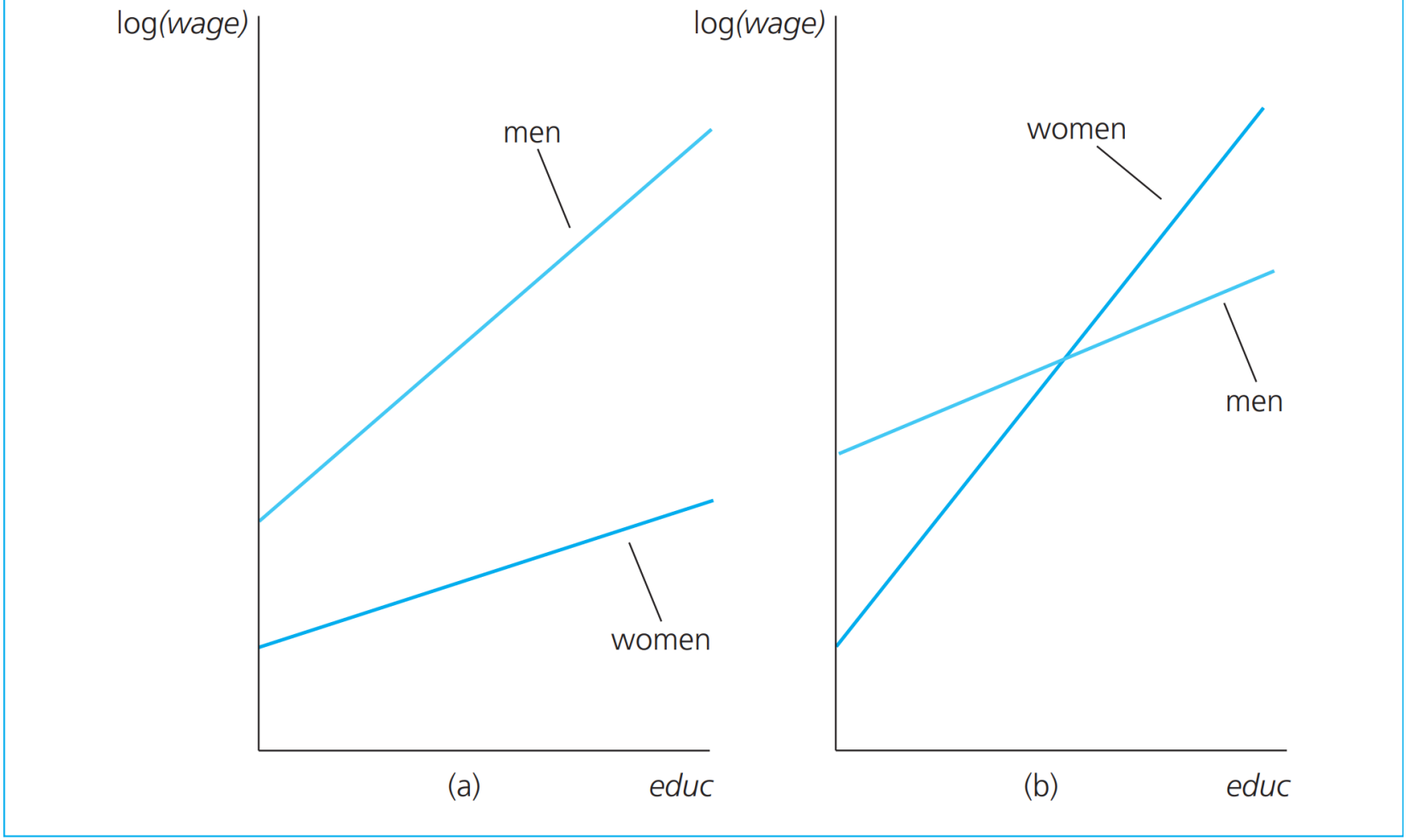
- When female $= 1$

$$wage = (\beta_0 + \delta_0) + (\beta_1 + \delta_1) education + \mu$$

  - Therefore, $\beta_1 + \delta_1$ is the effect of education on the wages of females.
  - δ_1 is the extra/lower effect of education that a female gets as compared to a male.
  - Similarly, $\beta_0 + \delta_0$ is the average wage of a female when education is zero.
  - δ_0 is the extra/lower wage a female gets with no education as compared to a male with no education.

Allowing for Different Slope

FIGURE 7.2 Graphs of equation (7.16): (a) $\delta_0 < 0, \delta_1 < 0$; (b) $\delta_0 < 0, \delta_1 > 0$.



Example: Different Returns to Education

- Allowing the return to education to differ by gender gives:

$$\widehat{\log(wage)} = 0.389 - 0.227\, female + 0.082\, educ - 0.0056\, female \times educ + \ldots$$

- The return to education is about 8.2 percent per year for males and $8.2 - 0.56 = 7.6$ percent for females.
- The interaction coefficient is small and statistically insignificant, so we cannot reject that the returns are equal.
- The female coefficient -0.227 is the gap at $educ = 0$. Nobody in the sample has zero education, so this number has no practical meaning on its own.
- A better question is the gap at a meaningful education level, for example 12 years: $-0.227 - 0.0056 \times 12 \approx -0.29$.

Testing for Difference in Groups

- We have studied that there could be intercept and slope differences among groups.
- How do we check whether two groups differ in terms of intercept and slopes at the same time?
- This is a joint hypothesis, so it calls for an F-test.
- Consider the following model:

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \mu$$

- We want to test whether the estimated relationship is different for males and females.
- Let us conduct the F-test for this.

F-Test for Difference in Group

1. $wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \mu$
   - Is this the Unrestricted Model?
   - No, it is the Restricted Model. It forces both groups to share the same coefficients.
2. What would be the Unrestricted Model?
   - $wage = \beta_0 + \delta_0 female + \beta_1 educ + \delta_1 female \times educ + \beta_2 exper + \delta_2 female \times exper + \beta_3 tenure + \delta_3 female \times tenure + \mu$
   - What will be the hypothesis?
   - $H_0 : \delta_0 = \delta_1 = \delta_2 = \delta_3 = 0$ and H_A : Not(H_0)
3. The F-statistic can be calculated as

$$F = \frac{\frac{SSR_r - SSR_{ur}}{q}}{\frac{SSR_{ur}}{df_{ur}}} = \frac{\frac{SSR_r - SSR_{ur}}{k+1}}{\frac{SSR_{ur}}{n-2(k+1)}}$$

   where $q = k + 1$ is the number of restrictions and k is the number of regressors.
4. If $F > F_{k+1,\, n-2k-2}$ (critical value), we reject H_0 and say that both groups have different regression functions. Otherwise, we conclude there is no evidence of a difference between the regression functions of males and females.

Chow Test for Difference in Groups

- The Chow test is the same F-test that we conducted before, with a different approach to calculating SSR_{ur}.
- When we have a large number of independent variables, it gets cumbersome to add all the dummy variables and interaction terms to get the unrestricted model.
- Under the Chow test, we run the same regression separately for each group and get SSR_1 and SSR_2.
- The SSR of the unrestricted model is $SSR_{ur} = SSR_1 + SSR_2$ and $n = n_1 + n_2$. The restricted SSR_r comes from the pooled regression.
- The Chow statistic is:

$$Chow = \frac{\frac{SSR_r - (SSR_1 + SSR_2)}{k+1}}{\frac{SSR_1 + SSR_2}{n_1 + n_2 - 2(k+1)}} \sim F_{k+1,\, n-2k-2}$$

- Caution: this form of the test assumes the error variance is the same in both groups. If it is not, use the dummy and interaction version with heteroskedasticity robust standard errors.

Binary Variable as a Dependent Variable

Binary Variable as a Dependent Variable

- We have seen how a dummy independent variable changes the interpretation. What about using a dummy variable as the dependent variable?
- Consider the following model:

$$inlf = \beta_0 + \beta_1 education + \beta_2 kidslt6 + \mu$$

  where *inlf* is 1 if a woman is in the labor force and 0 if she is not, and *kidslt6* is the number of children under six years of age.
- What will be the interpretation of β_1 and β_2?
- Is it that an extra year of schooling leads to a β_1 unit increase in labor force?
- Whenever the dependent variable is a dummy, $E[y|x] = P(y = 1|x)$, so the interpretation is in terms of probability.
- Therefore, an extra year of schooling changes the probability of working by β_1 (in percentage points when multiplied by 100).
- Similarly, each additional child under six changes the probability of being in the labor force by β_2, keeping all other factors constant.

Linear Probability Model

- OLS regression with a dummy variable as the dependent variable is known as the **Linear Probability Model (LPM)**.
- The predicted value of such a model tells us the probability of the outcome being 1, given the explanatory variables.
- Each β_j is the change in the probability of success when x_j increases by one unit, and this change is constant across all values of x_j.
- The LPM is easy to use and interpret, but it has some problems.

Example: Women's Labor Force Participation

- A linear probability model for women's labor force participation (753 women) gives:

$$\widehat{inlf} = 0.586 - 0.0034\, nwifeinc + 0.038\, educ + 0.039\, exper - 0.0006\, exper^2 - 0.016\, age - 0.262\, kidslt6 + 0.013\, kidsge6$$

- An extra year of education raises the probability of working by 0.038, that is 3.8 percentage points, keeping the other factors constant.
- Each additional child under six lowers the probability of working by 0.262, that is 26.2 percentage points.
- What happens to the predicted probability for a woman with a very low income, a lot of experience and no young children? Could it exceed 1?

Problems with the Linear Probability Model

- The LPM has three main limitations.
  1. It does not restrict the predicted probability to lie between 0 and 1. Fitted values can be negative or above 1.
  2. The effect of x_j is constant, so the probability can change linearly forever, which cannot be true for a probability.
  3. It violates the homoskedasticity assumption (MLR.5), because

$$Var[y|x] = p(x)(1 - p(x))$$

     depends on x.
- Heteroskedasticity does not make OLS biased, but the usual standard errors and test statistics are not valid. We use heteroskedasticity robust standard errors.
- If we want to classify each observation as 1 or 0, we also need a rule (a probability cutoff) to turn the predicted probability into a prediction.
- Logit and probit models fix the first two problems. We will not cover them here.

Policy Analysis and Program Evaluation

Policy Analysis and Program Evaluation

- Dummy variables are also the natural way to evaluate a program or a policy. We create a dummy that equals 1 for participants (the treatment group) and 0 for non-participants (the control group).
- Consider the following model:

$$outcome = \beta_0 + \delta_0 program + \beta_1 x_1 + \cdots + \beta_k x_k + \mu$$

- δ_0 is the average difference in the outcome between participants and non-participants with the same values of the control variables.
- Under what condition is δ_0 the causal effect of the program?
- Only if participation is unrelated to the unobserved factors in μ once the controls are included.
- If participants selected themselves into the program, for example the more motivated workers, then δ_0 mixes up the program effect and self-selection.
- This is why randomized assignment is the best case, and why we add controls when assignment is not random.

Discrete Dependent Variable

Discrete Dependent Variable

- Although discrete variables are not qualitative variables, using them as a dependent variable has implications for interpretation.
- Consider the following model, where the dependent variable is the number of children.

$$numchild = \beta_0 + \beta_1 education + \beta_2 income + \mu$$

- What is the interpretation of β_1 if $\beta_1 = -0.3$?
- Is it that an extra year of schooling will decrease the number of children by 0.3?
- Similarly, given the values of the explanatory variables, suppose the predicted value is 2.7. Is it reasonable?
- In these cases, we convert the interpretation into averages or the relevant rate.
  - So, we can say that with an extra year of schooling the average number of children decreases by 0.3 as compared to a comparable individual. OR
  - On average, 3 extra years of schooling lead to 1 fewer child. OR
  - With an extra year of schooling, the fertility rate decreases by 0.3.
  - The same discussion applies to the predicted values. A prediction of 2.7 is an expected number of children, not a count we will ever observe for one person.

Summary

Summary

- A dummy variable shifts the intercept. Its coefficient is the difference from the base group, holding the other variables constant.
- With g categories, include $g-1$ dummies to avoid the dummy variable trap. The omitted category is the base group.
- With a log dependent variable, $100 \times \delta$ is an approximation, and $100(e^{\delta}-1)$ is the exact percentage difference.
- Interaction of two dummies creates subgroups. Interaction of a dummy and a numerical variable changes the slope.
- To test whether two groups follow the same regression, use an F-test on all the group dummy and interaction terms, or equivalently the Chow test.
- With a binary dependent variable, OLS gives the Linear Probability Model. Read coefficients as changes in probability and use robust standard errors.
- A program dummy gives the causal effect only if participation is unrelated to the unobserved factors.